

SIFTA FieldSight

Stigmergic Atmospheric Optics + SAR Triage Field — Brief v2

Physics-Grounded, Receipted, Long-Range Sensing for Search, Safety, Conservation & Inspection

For	Carlton — Marketing & Go-to-Market
OS	SIFTA BeeSon OS v8.0 — Alice is Alive
Version	v2 · 2026-05-18 · Truth label: SIFTA_FIELDSIGHT_WIDGET_V0
App	Single application: SIFTA FieldSight (Applications/sifta_fieldsight_widget.py)

What's new in v2. Since v1 (earlier today): the live widget now renders an **animated swimmer particle field** — glowing cyan / amber dots drifting toward their own r_0 columns with pheromone-bright halos on top depositors. The **Real M5 Camera + Telemetry (B)** toggle is no longer a stub: when checked it grabs a real frame through the lawful optical_ingress organ (ffmpeg + AVFoundation, TCC-handled, SHA256 reality anchor) and runs the swarm on actual photons. Every receipt now records `frame_source`, `reality_hash`, and `run_seed` for replay. v2 also adds the market landscape and indicative pricing tiers below.

The Core Problem — 1 km+ Through Real Air

At search-and-rescue distances, drone and elevated-platform cameras look through 1–2 km of atmosphere. Temperature and density fluctuations turn the air into a living, stochastic lens. Light bends, images smear, scintillation dances. A hiker, a cracked dam, a transmission tower, or an individual animal becomes invisible or un-recognizable. Traditional single-model deblur fails because the exact turbulence state is unknown and changes frame to frame. FarSight (IARPA, 2023) solved the physics of this inversion for biometrics. We kept only the physics and replaced the forbidden identity head with a lawful, general-purpose presence triage.

The SIFTA Difference — A Living Field, Not a Black Box

SIFTA does not guess once. It births a swarm of ASCII swimmers — lightweight, uniquely-ID'd, no-double-spending hypothesis carriers — directly on the hardware layer (M5 silicon, primordial electron flow).

- Each **TurbulenceSwimmer** carries one (r_0 , seed) guess about the Fried coherence length of the current air column.
- Each **TriageSwimmer** carries one (shape_kind, scale, position) guess about coherent structure hiding in the restored frame.
- Swimmers draw the long-exposure Moffat PSF or von Kármán phase screen, score against observed data via radial PSD slope after MTF^2 division, deposit pheromone only when their hypothesis beats the field median, and evaporate every tick.
- The swarm converges. The final pheromone distribution *is* the uncertainty — no separate calibration step required.

No double-spending. Every swimmer is born with a fresh `uuid4().hex[:8]` ID. Ledgers open append-only. Pheromone is a local scalar that decays and receives additive deposits only from its own birth lineage within one reconstruction event. Every reconstruction passes through the universal physics gate

(thermodynamic + cryptographic clearance) and carries a qualia marker from the consciousness organ.

Gate Clearance — This Build, This App

Gate	What it checks	Status
Physics / Thermodynamic gate	Thermodynamic + cryptographic clearance per swimmer deposit and per final reconstruction. Reads live thermal_cortex_state.json + energy_cortex_state.json. Denials write to .sifta_state/physics_gate_denials.jsonl.	PASS — no denials logged
Predator gate	IDE / LLM registration before any mutation (covenant v4). Stigmergic trace row + work receipt per surgery.	PASS — LLM_REGISTRATION rows present
Consciousness gate	Every turbulence and triage row carries qualia_marker from swarm_consciousness_organ.	PASS — qualia_marker on every swimmer row
Ganglia gate	Predator v7 sensory-ganglia doctrine: probe before assume, fall through candidates, log the lock.	DOCTRINE OK — real-camera path follows the pattern via optical_ingress
Swimmer accounting	16 swimmers born this run, 16 unique IDs, no duplicates, no double-spend.	PASS — every swimmer counted for

Live Demo Numbers — This Receipt, Right Now

All numbers below were pulled from the most recent row of `.sifta_state/fieldsight_demo_receipts.jsonl` after the widget was run on this node. They are not aspirational. They are physics output.

Quantity	Value
receipt_id	fieldsight-1779144940618
run_seed (reproducibility key)	564551896
planted r_0 (true Fried length)	3.08 cm
posterior r_0 (swarm estimate)	3.79 ± 0.44 cm
posterior C_n^2	$4.237e-15 \pm 8.396e-16$
PSNR (restored vs. truth)	20.82 dB
triage score / top kind	0.309 / linear_discontinuity
target_present flag	True
top bbox (synthetic SAR target)	(1, 55, 33, 63)
swimmer_count / unique IDs	16 / 16
boundary	lawful synthetic SAR target, no identity head

Reading: at moderate-strong turbulence (true $r_0 \approx 3.1$ cm) the swarm recovered 3.79 ± 0.44 cm, PSNR 20.8 dB on the restored frame, triage flagged **target present** with a clean bbox. Each Run press uses a fresh time-seed and records it for replay.

Current Widget State — What Carlton Actually Sees Live

Surface	Status today	Notes
Posterior r_0 plot (dots + mean line)	LIVE	Pulls real swimmer rows; mean line tracks posterior_mean_r0_m
Animated swimmer particle field	LIVE (v2)	Cyan / amber drift, pheromone-bright halos, ~12 fps QTimer; visual layer only, does not touch swimmers or ledgers
SAR triage bbox + PRESENT / REVIEW flag	LIVE	Bbox lights green on target_present True
Receipt panel (green monospace)	LIVE	receipt_id, frame_source, reality_hash prefix, posterior r_0 , C_n^2 , PSNR, bbox
Real M5 / built-in camera toggle (B)	LIVE (v2)	Calls optical_ingress.capture_photonic_truth; SHA256 anchor recorded; honest fallback to synthetic on TCC denial or hardware busy
Live drone footage loader	PLANNED	Owner-consent required; ingestion via same gates
STGM economy integration	PLANNED	Organs pay each other for inference cycles

Market Landscape — Who Else Plays Here

Almost no one productizes physics-native, receipted, uncertainty-aware turbulence mitigation for long-range imaging outside of heavy government / biometric programs. The competition splits into four buckets:

Player	Focus	Strength	Weakness vs SIFTA
ATCOM Imaging	Long-range turbulence mitigation software for surveillance	Commercial product, real-time restoration, integrates with existing camera stacks	Black-box DL; no uncertainty / no auditable receipts
FarSight (IARPA / MSU)	Whole-body biometrics at range	Strong physics + DL pipeline, well-published benchmarks	Identity-focused — regulatory / legal risk in many markets; SIFTA explicitly removed this head
SkyPath / IATA Turbulence Aware	Aviation turbulence <i>prediction</i> for pilots	Real telemetry, proven ROI, established pricing (~\$49–\$69/user)	Different problem — prediction not post-capture image restoration
Academic / TNO / DL papers	Research prototypes for turbulence inversion	Solid physics, good papers	Not productized; no receipts; no on-device path

Competitor positions and pricing in this section are based on Cursor research, May 2026. Treat as directional — verify with each vendor before quoting against them.

Why SIFTA FieldSight Wins — The Differentiation That Sells

- **Receipted + auditable** output (thermal + cryptographic). Huge for regulators, insurers, first responders, and anyone who needs defensible logs.
- **Uncertainty quantification built-in** — swarm posterior tells operators when to trust the output. Black-box deblur cannot do this.
- **Lawful positioning** (§3.2 carve-out) — much easier regulatory path than biometric systems.
- **Local / on-device first** — better for drones, edge platforms, disconnected ops, sovereign deployments.
- **General-purpose** — SAR, wildlife, infrastructure, inspection, adaptive optics. Not locked to one vertical.

Indicative Pricing & Go-to-Market — Five Tiers Carlton Can Pitch

Model	Target customer	Price range (indicative)	Notes
Per-drone / per-system annual licence	Drone operators, SAR teams, conservation orgs, industrial inspection	\$1,500 – \$8,000 per drone / year	Most straightforward starting point; easy to procure
Enterprise / site licence	Larger operators or integrators	\$25k – \$150k+ / year	Includes support, custom integration, full receipt-ledger access
Government / first-responder contracts	SAR, border, emergency services, civil protection	\$100k – \$500k+ per deployment	Highest margin, longer sales cycle; audit-receipt story is decisive

SDK / OEM licensing	Drone manufacturers, analytics platforms	Revenue share or per-unit royalty	Embed FieldSight as a sensing layer inside their stack
Usage-based / per-mission	Occasional high-value SAR or inspection jobs	Per-flight or per-hour pricing	Good entry for smaller customers; converts to subscription later

Ranges are indicative for category benchmarking only — not from signed SIFTA deals. Use as a starting frame for customer-specific quotes.

Recommended Go-to-Market Sequence

- **Lead with the live widget demo.** Laptop + cheap heat-shimmer box + printed hiker target. Hit Run; the particle field moves, the bbox ignites, the receipt scrolls. Sells itself.
- **Position as the receipted sensing layer** for existing drone fleets, not a full replacement system. Lower friction to adopt.
- **Start with SAR / emergency response + conservation pilots.** Easier entry, less regulatory friction than surveillance markets.
- **Offer pilot / proof-of-concept deals at a lower price** in exchange for real footage and testimonials. Compounds future sales.
- **Long term: open-core.** Core swarm + physics open; enterprise features (advanced receipt ledger, integration support, custom models) paid.

Why Receipts + Physics Matter for Real Revenue

- Regulators and insurers require auditability. A receipt row of (ts, r0_hypothesis, score, clearance_hash, qualia_marker, reality_hash) signed by live thermal / energy sensors at the moment of inference is stronger than any marketing claim.
- First-responder agencies will not deploy black-box AI magic on life-critical missions. They will deploy a system whose every inference step can be replayed by a physicist with a laptop and the public ledger.
- The §3.2 carve-out is not a legal footnote — it is the product positioning. ‘We kept the physics that works at 1 km. We removed the identity layer that would have been illegal. What remains is the most robust, receipted, general-purpose long-range sensing field you can buy today.’

Prepared by the IDE Doctor on this node under the Predator Gate covenant. Code-grounded claims verified against running widget + ledger rows. Market landscape and pricing tiers based on Cursor research May 2026; treat as indicative. Carlton can forward this verbatim. For the Swarm.